

AgentBoard: An Analytical Evaluation Board of Multi-turn LLM Agents

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Chang Ma et al. (The University of Hong Kong) AgentBoard: An Analytical Evaluation... May 24, 2026 1 / 27

Background: LLM Agents are Transforming Decision-Making

- LLMs are evolving from chatbots to agents that perceive, plan, and act.
- Applications span embodied robotics, web automation, games, and tool use.
- Agent evaluation must reflect real-world constraints: partial observability and multi-turn interaction.

Chang Ma et al. (The University of Hong Kong) AgentBoard: An Analytical Evaluation... May 24, 2026 2 / 27

The Problem: Measuring Agent Competence in Realistic Settings

- Current benchmarks are single-turn or fully observable—poor proxies for agency.
- Reporting only final success masks meaningful intermediate progress and skill gaps.
- Lack of unified, cross-domain, multi-turn evaluation limits fair comparison and diagnosis.

Chang Ma et al. (The University of Hong Kong) AgentBoard: An Analytical Evaluation... May 24, 2026 3 / 27

Core Idea: AgentBoard for Analytical, Multi-Turn Agent Evaluation

- Unified benchmark and open-source framework across 9 partially-observable environments.
- Fine-grained progress rate to capture incremental advancement per step.
- Interactive toolkit to analyze sub-skills, grounding, difficulty, and long-range behavior.

They show a unified, partially-observable, multi-round evaluation board with per-step analytics.

Chang Ma et al. (The University of Hong Kong) AgentBoard: An Analytical Evaluation... May 24, 2026 4 / 27

- Text-in/text-out interface standardizes multi-round interaction and feedback.
- Partially observable environments with deterministic transitions and error handling.
- Supports consistent action formatting and cross-domain comparability.



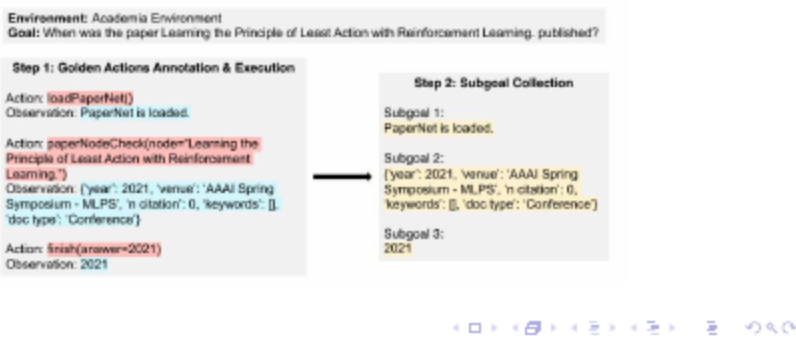
- Reflex agent with sliding-window memory to mitigate context limits.
- Act-only prompting as default; ReAct supported but optional to avoid confounders.
- Consistent one-shot examples per task for fair comparison.

- Tracks maximum attained progress over steps to form a monotonic trajectory.
- Two implementations: r_{match} (continuous similarity) and $r_{subgoal}$ (discrete completion).
- Enables per-step diagnosis, difficulty-aware analysis, and fair cross-task aggregation.
- Example: In PDDL, satisfying 3 of 5 goal predicates yields $r_{match} = 0.6$; in WebShop, adding the correct item before checkout raises $r_{subgoal}$ (e.g., 0.4 $\rightarrow$ 0.6) even if the purchase fails.

- Applies when intermediate states are directly comparable to the goal state.
- Computes a matching function to measure proximity to completion.
- Used for settings like PDDL/Web tasks with well-defined state constraints.

- Unique, human-verified sequences of necessary subgoals per task.
- Regex-based checking over observations for automatic subgoal completion.
- Ensures robustness in tasks with ambiguous intermediate states (e.g., embodied, web).

They illustrate unique, human-verified subgoal sequences and automatic, regex-based completion checks.

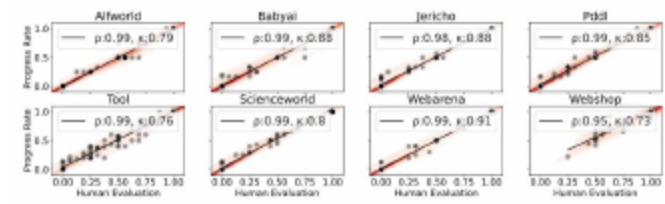


Task Suite: Nine Environments, Four Scenario Families

- Embodied: AlfWorld, ScienceWorld, BabyAI.
- Games: Jericho, PDDL domains; Web: WebShop, WebArena.
- Tools: Tool-Query (APIs), Tool-Operation (Todo, Sheets).

- User study: progress rate correlates with human judgment ($\rho > 0.95$).
- High inter-annotator agreement supports reliability.
- Enables scalable, automated yet trustworthy evaluation.

They show that the progress metric aligns with human judgment across tasks, supporting automated, trustworthy evaluation.



Environment Adaptations for Analytical Evaluation

- The authors textualized interfaces (e.g., BabyAI, PDDL) for consistent action/observation formats.
- The authors re-annotated subgoals (e.g., ScienceWorld) to enforce unique sequences where needed.
- The authors revised web scoring (WebShop/WebArena) for continuous progress tracking.

- They provide an interactive WandB panel to compare models, inspect trajectories, and visualize sub-skill radar.
- Difficulty breakdowns reveal compositionality limits and bottlenecks.
- Grounding analysis (valid-action rates) separates planning from execution errors.

They show per-task and per-model panels that localize bottlenecks at trajectory level.



Main Results Across Tasks

- Progress rate is more discriminative than success, unveiling hidden capability gaps.
- Proprietary models lead; coding- and agent-tuned models show advantages.
- Long-range interaction remains a bottleneck for many open-weight models.

Table: Table 3: Performance of different LLMs sorted by success rate.

Model	Embodied AI			Game		Web		Tool		Avg.
	ALF	SW	BA	JC	PL	WS	WA	TQ	TO	
GPT-4		65.5/43.3	78.8/52.2	70.7/56.2	52.4/35.0	81.2/61.7	76.5/39.0	39.4/15.1	85.1/68.3	80.8/60.0
Claude2		34.1/24.6	32.0/11.1	48.1/37.5	20.4/0.0	61.4/40.0	74.6/37.8	36.4/8.6	73.5/48.3	70.0/47.9
Gemini1.5-Flash ‡		40.9/15.7	17.8/ 4.4	50.6/38.4	11.1/0.0	23.5/6.7	72.3/24.7	32.4/11.1	73.9/46.7	59.6/27.5
Claude3-Haiku	20.7/ 2.2	43.7/14.4	34.4/24.1	34.7/10.0	31.9/13.3	73.6/30.3	26.8/12.7	61.9/24.0	62.4/27.5	48.9/26.2
Llama3-70b		29.6/12.7	30.4/ 7.8	41.1/27.7	16.0/5.0	32.2/20.0	74.6/29.9	35.6/12.6	52.6/36.7	68.6/37.5
GPT-3.5-Turbo		35.6/17.2	31.9/18.9	51.7/39.3	19.9/5.0	25.0/5.0	76.4/35.1	25.5/4.6	69.4/45.0	43.5/20.6
xLAM-70b		53.4/42.5	15.4/ 1.1	37.7/28.6	16.2/5.0	38.4/16.7	73.6/32.7	34.5/11.3	66.5/38.3	43.3/17.6
DeepSeek-67b		34.5/20.9	36.1/10.0	31.7/22.3	13.7/0.0	22.0/6.7	72.7/31.9	23.9/5.7	71.4/40.0	65.2/30.0
Text-Davinci-003	18.8/ 9.0	28.9/ 7.8	17.5/14.3	28.6/10.0	31.7/11.7	72.3/29.5	16.2/2.5	65.0/38.3	56.2/22.5	41.9/20.2
GPT-3.5-Turbo-16k	25.2/ 4.5		2.2/ 0.0	45.1/33.9	16.1/0.0	22.6/3.3	73.8/27.9	23.7/6.1	59.1/31.7	37.2/7.5
AgentLM-70b		58.4/50.7	13.0/ 1.1	38.0/27.7	8.8/0.0	13.0/3.3	72.0/31.1	13.0/5.3	50.5/13.3	41.4/19.7
Lemur-70b	10.8/ 0.7	33.4/ 5.6	19.4/9.8	10.1/0.0	9.7/3.3	71.8/11.6	12.2/3.3	72.0/28.3	37.7/12.5	26.5/7.5
										40.2/20.4
										40.5/17.5
										38.5/17.2
										37.2/16.2
										34.2/13.6
										31.8/0.0
										33.3/14.7
										30.8/8.3
										28.0/7.7

- They evaluate with one-shot in-context examples per task and a unified reflex-agent policy.
- Models include proprietary (GPT-4, Claude, Gemini), open-weight (Llama, Mistral, DeepSeek), and agent-tuned (xLAM, AgentLM).
- Metrics: Progress rate (primary) and success rate (complementary).

Note: Models and full baseline citations are listed in “Baseline Methods: Sources.”

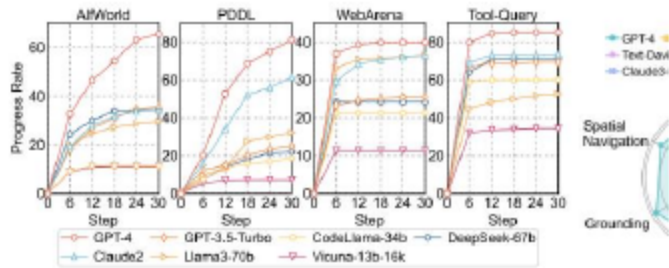
Most Surprising Finding

- They show that success rate dramatically understates agent capabilities in hard, low-success regimes: several models make 40–60% progress despite near-zero success.
- Progress-rate reorders models in non-trivial ways: a code-pretrained 34B model surpasses a larger 70B general LLM on progress in multiple tasks.
- On web/tool tasks, high grounding (valid actions) coexists with low success—indicating reasoning/compositional planning, not tool use, is the limiting factor.

Sustaining Progress Over Long Interactions

- Proprietary models maintain progress over 30+ steps in embodied/planning tasks.
- Many open-weight models plateau after ~6 steps—memory/planning limits.
- Progress trajectories illuminate when and where agents stall.

They show proprietary models sustain growth past 30 steps, while most open-weight models plateau near step ~6.



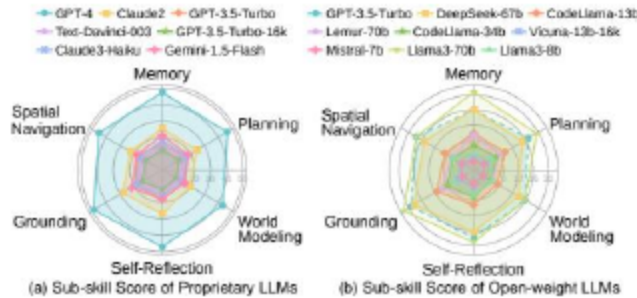
Grounding Analysis: Valid-Action Rates

- Across tasks, proprietary models issue more valid actions.
- Yet valid-action leadership does not guarantee success; some models remain bottlenecked by planning/composition.
- They show grounding accuracy helps disentangle execution slips from reasoning gaps.

Difficulty and Sub-skill Profiles

- All models drop sharply on hard cases (more subgoals/constraints).
- Sub-skill radar: GPT-4 leads in memory, planning, world modeling, reflection, grounding, navigation.
- Grounding and compositional planning remain key challenges for many models.

They compute sub-skill scores from rubric-based tags over trajectory steps (memory, planning, world modeling, self-reflection, grounding, navigation).



Positioning: How AgentBoard Differs from Prior Benchmarks

- Unifies task diversity, multi-round interaction, partial observability, and fine-grained metrics.
- Adds analytical tooling for sub-skill diagnosis and trajectory-level insights.
- Enables consistent, cross-domain, multi-turn comparisons under one protocol.
- Compared to AgentBench (Liu et al., 2023) and MINT (Wang et al., 2023), they unify partial observability, multi-round interaction, and fine-grained progress for analytical diagnosis.
- They provide trajectory-level tooling rarely present in prior work.

Ablation: Handling Long Context in Multi-Turn Agents

- Sliding-window memory consistently outperforms cutoff and summary across embodied, web, game, and tool tasks.
- The choice of memory strategy materially shifts both progress trajectories and final outcomes.
- They show summary/cutoff variants often plateau earlier on long-horizon problems.

Qualitative Examples and Case Studies

- Embodied (BabyAI; medium): Steady $r_{subgoal}$ gains via correct key acquisition and door navigation; failures from mis-sequenced objectives.
- Game (Jericho; hard): Correct world-model updates but invalid action formatting causes stalls—grounding vs. reasoning decoupling.
- Tool (Todo/Sheets; easy): Strong grounding on CRUD operations; cross-app composition errors under multi-step coordination.
- Embodied (AlfWorld; hard): $r_{subgoal}$ rises 0.20→0.40 after correct object discovery, then stalls after erroneous container search—grounding intact; exploration/planning insufficient.
- WebShop (medium): Valid tool calls add the correct item ($r_{subgoal} \approx 0.50$) but checkout fails due to missing coupon application—composition across steps is the blocker.



Practical Insights for Building Stronger Agents

- They find that coding-oriented pretraining and agent-tuning provide measurable gains.
- Grounding improvements and planning scaffolds help on hard, compositional tasks.
- Practitioners can use progress rate to track gains when success is still rare.

Conclusion & Outlook

- They conclude AgentBoard standardizes cross-domain, multi-turn, partially observable agent evaluation and provides a validated progress rate with rich analytics.
- They plan to automate subgoal generation with stronger LLMs and expand to dynamic, real-world, multimodal settings with safety guardrails.
- Adoption: use AgentBoard for deployment readiness checks and to track progress when success is rare.

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